Flexibility of Nuclear Power in The Nordic Electricity Market

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Abstract

Introduction Although more than 50% of Nordic electricity supply is from hydropower, nuclear power is also an important source in about 20% of the generation mix. However, with increased electricity demand foreseen in adjacent sectors, such as heat, industrial processes, and transport, capacity expansion of variable renewable energy (VRE), such as wind and solar, will be likely in the near future. While levelised costs of VRE have been declining, their output is still hampered by intermittency, which necessitates variability management. Hydropower has traditionally been a source of such flexibility, but nuclear plants could also operate in load-following mode. Equilibrium models used to analyse electricity markets typically treat nuclear plants as generic thermal technologies that operate within given capacity limits and ramp rates. Yet, nuclear plants' flexibility depends on the fuel cycle of reactors, which can be modelled akin to hydro reservoirs.

Research Question In this thesis, an equilibrium model of the Nordic power sector is adapted to model nuclear plants' flexibility as a resource stock to be deployed over the year. The primary research question is: "How can modelling of nuclear fuel cycles as a resource stock deployed over the year impact the assessment of nuclear-power-plant flexibility and the overall equilibrium of the Nordic electricity market under high VRE penetration?"

Method A deterministic partial-equilibrium model of the Swedish power system is developed, covering the four Swedish bidding zones (SE1–SE4) at hourly resolution over a full year using 2018 as the base year. The model maximises social welfare under perfect competition, formulated as a quadratic program implemented in GAMS using the CPLEX solver. Two model variants are compared across VRE penetration scenarios representing 2030,

2040, and 2050 conditions: *Model R*, which represents nuclear plants using conventional ramp-rate constraints only, and *Model F*, which incorporates both a fuel-energy stock constraint and a cycling-operations budget. The shadow prices on these constraints are interpreted as the fuel value and flexibility value respectively, directly analogous to the water value in hydropower reservoir models.

Results

Discussion

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Abbreviations

CPLEX	IBM ILOG CPLEX Optimisation Studio
DSR	Design Science Research
ENTSO-E	European Network of Transmission System Operators for Electricity
EPR	European Pressurised Reactor
EU	European Union
GAMS	General Algebraic Modelling System
GW	Gigawatt
HVDC	High Voltage Direct Current
IAEA	International Atomic Energy Agency
IEA	International Energy Agency
IPCC	Intergovernmental Panel on Climate Change
KKT	Karush–Kuhn–Tucker
LMP	Locational Marginal Price
MW	Megawatt
MWh	Megawatt-hour
NEA	Nuclear Energy Agency
NTC	Net Transfer Capacity
OECD	Organisation for Economic Co-operation and Development
PWR	Pressurised Water Reactor
QCP	Quadratic Constrained Programme
QP	Quadratic Programme
SDDP	Stochastic Dual Dynamic Programming
SMR	Small Modular Reactor
TSO	Transmission System Operator
VRE	Variable Renewable Energy

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1

Introduction

1.1 Background

Climate change represents one of the most significant challenges facing human civilisation, with the energy sector serving as the dominant contributor to global greenhouse-gas emissions. The Intergovernmental Panel on Climate Change warns that limiting global warming to 1.5°C above pre-industrial levels requires rapid and far-reaching transitions in energy systems [23]. Within this context, electricity generation alone accounts for approximately 40% of energy-related CO_2 emissions, making power-sector decarbonisation an urgent policy priority worldwide [20]. The European Union has responded through the European Green Deal, which establishes legally binding targets for climate neutrality by 2050 and an intermediate goal of reducing emissions by 55% relative to 1990 levels by 2030 [12]. These targets necessitate fundamental changes in electricity generation, transmission, and consumption patterns across member states. Variable renewable energy (VRE) sources—principally wind and solar—have emerged as cornerstone technologies for this transition, with their levelised costs declining dramatically over the past decade [21].

Sweden presents a particularly compelling case study for power-system decarbonisation. The country already enjoys one of the world's cleanest electricity mixes, with hydropower supplying more than 40% of total generation and nuclear power contributing approximately 40% [10]. A distinctive feature of the Swedish system is its abundant hydropower resources: Norway and Sweden together account for approximately half of Europe's hydropower capacity, providing extensive reservoir storage that buffers variations in both demand and VRE generation [16]. Wind capacity has grown rapidly, particularly in northern Sweden, and further VRE expansion is projected under national and EU decarbonisation targets [12, 21].

Nuclear power is a cornerstone of Swedish electricity supply, providing reliable, low-carbon baseload generation. Sweden operates six reactors across three sites: Ringhals (2 units), Forsmark (3 units), and Oskarshamn (1 unit), with a combined installed capacity of approximately 7.5 GW [19]. As VRE penetration increases, the operational role of these plants is evolving from traditional baseload operation toward potential load-following modes that adjust output to accommodate renewable fluctuations [30]. This shift raises important questions about the technical and economic limits of nuclear flexibility in a system where hydropower has historically served as the primary balancing resource [16].

1.2 Problem Statement

Equilibrium models used to analyse electricity markets typically represent nuclear power plants as generic thermal technologies characterised by maximum capacity limits, constant marginal operating costs, and ramping constraints [34, 32]. This representation implicitly assumes that nuclear fuel is effectively infinite within the modelling horizon—that generating today imposes no reduction in the ability to generate tomorrow, aside from ramp-rate limitations. More precisely, it assumes that the *availability of fuel cycles is unlimited*, so that the intertemporal allocation of finite fuel is never a binding constraint.

This assumption does not reflect the physical reality of nuclear fuel consumption. A nuclear reactor core contains a fixed amount of fissile material that is gradually depleted during operation. After 12–24 months of continuous operation the fuel must be replaced during a refuelling outage lasting several weeks [18]. Consequently, nuclear plants face an intertemporal trade off conceptually identical to that faced by hydro reservoirs: generating today consumes resources that cannot be used tomorrow [7]. The key difference is that hydro reservoirs receive natural inflows replenishing the water stock, whereas nuclear fuel receives no such replenishment until the next refuelling outage, making nuclear fuel management a pure depletion problem.

This oversight has significant implications for system planning. If conventional models overestimate nuclear flexibility by ignoring fuel constraints, then they may underestimate the costs of VRE integration and overstate the ability of existing nuclear fleets to accommodate high VRE penetrations. Recent work by Blanchard and Massol [3] has begun addressing this gap by modelling nuclear flexibility as a stock constraint on cycling operations within a stochastic dynamic programming framework, estimating the marginal value of flexibility for the French nuclear fleet. However, the fuel-energy constraint—the intertemporal allocation of finite fuel across periods—remains largely unexamined in perfect-competition equilibrium models. Accurate representation of nuclear fuel cycles is therefore essential for informed policy and investment decisions.

It is important to distinguish between the contribution of Blanchard and Massol [3] and the perspective adopted in this thesis. Blanchard and Massol focus on the French nuclear fleet of 56 reactors, operating in a system with limited hydropower storage, and model the cycling-operations constraint using stochastic dynamic programming. This thesis, by contrast, examines the Swedish context—six reactors co-located with vast hydropower reservoir resources—and incorporates *both* the cycling constraint and the fuel-energy constraint within a deterministic, partial-equilibrium framework. The presence of large hydro reservoirs in the Swedish system creates a fundamentally different flexibility landscape: hydropower has historically absorbed most balancing responsibility, and the question of how nuclear fuel-cycle constraints interact with hydro scheduling under high VRE penetration has not been studied in the existing literature.

Accordingly, this research problem belongs to an interdisciplinary area that combines the subjects of computer and systems sciences (decision and risk analysis/operations research/management science) and energy economics.

1.3 Research Question

This thesis addresses the following primary research question:

> *How does modeling nuclear fuel cycles as a resource stock, rather than using ramp rates, affect the assessment of nuclear flexibility and the overall equilibrium of the Swedish electricity market under high VRE penetration?*

The question is decomposed into two sub-questions:

1. *How can the operational constraints of a nuclear fuel cycle be modeled within a deterministic, partial-equilibrium framework of the Swedish power system, analogous to a hydropower reservoir?*
2. *Under high VRE scenarios, what is the comparative value of nuclear flexibility modeled via fuel cycles versus conventional models that use only ramp rates, measured in terms of system cost, market prices, variable renewable energy curtailment, and social welfare?*

1.4 Delimitations

This study focuses on the Swedish power system, represented through Sweden's four bidding zones (SE1–SE4). The temporal scope encompasses one-year simulations with hourly resolution using 2018 as the base year, with future scenarios targeting 2030, 2040, and 2050 time horizons based on ENTSO-E projections [11].

Technologies included comprise nuclear power with fuel-cycle constraints, hydropower with reservoir dynamics and natural inflows, wind and solar VRE with hourly availability profiles, thermal backup generation, and a zonal transmission network. The market structure assumes perfect competition within a welfare-maximising equilibrium framework following Chen et al. [1]. Market power and strategic behaviour are not considered.

The model is deterministic and does not include stochastic uncertainty. Nuclear refuelling outages are not explicitly modelled as discrete events; fuel is instead tracked continuously over the operating cycle. Only existing technologies are considered, with no endogenous capacity expansion. Demand is price responsive with linear inverse demand functions, and exogenous imports from outside Sweden are excluded.

1.5 Structure of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 provides a comprehensive review of the theoretical foundations underlying this research, including equilibrium models of electricity markets, the impact of increased VRE adoption on dispatchable plant operations, the representation of nuclear plant flexibility via ramp rates versus fuel cycles, and the research gap this thesis addresses. Chapter 3 describes the research strategy, data collection methods, and mathematical formulation of the equilibrium model, including the novel nuclear fuel-cycle constraints and implementation in GAMS using the CPLEX solver. Chapter 4 presents the outcomes of model simulations under various VRE penetration scenarios, comparing the conventional ramp-rate representation against the novel fuel-cycle representation. Chapter 5 interprets the results in the context of existing literature, explores policy implications, acknowledges limitations, and suggests future research directions, incorporating the key findings and conclusions of the thesis.

2

Extended Background

This chapter provides an extended background on the impact of increased VRE adoptions and the representation of nuclear plant flexibility. Following that, an overview of studies that use mathematical models to support decision analysis regarding welfare maximization under perfect competition will be surveyed since this thesis finds its root in the decision and risk analysis and addresses the difference in conventional model and our study.

2.1 Equilibrium Models of Electricity Markets

The analytical foundation of this thesis is the welfare-maximising equilibrium model of an electricity market. Schweppe et al. [32] established the canonical framework in which a system operator maximises the sum of consumer and producer surplus subject to network and technical constraints. The dual variables of the nodal power-balance constraints yield *locational marginal prices* (LMPs)—the marginal cost of serving one additional unit of load at each location—which serve as the economic signals for dispatch, consumption, and transmission investment. Under perfect competition this centralised solution is equivalent to the outcome of decentralised profit- and utility-maximising behaviour, a result that justifies using welfare maximisation as the modelling framework [32].

Ventosa et al. [34] survey how this framework has been extended to study market design, transmission congestion, and strategic behaviour, while Gabriel et al. [14] provide a comprehensive treatment of complementarity methods for energy markets. This thesis follows the partial-equilibrium formulation of Chen et al. [1], in which the market-clearing problem is cast as a quadratic program with linear constraints: inverse demand functions are linear, making the consumer-surplus term quadratic, while all generation cost and network constraint expressions remain linear. The resulting problem is strictly convex, guaranteeing a unique solution and ensuring that the KKT conditions are necessary and sufficient for optimality [2]. Market power is set aside throughout, since the focus is on isolating the effect of the nuclear fuel-cycle representation on equilibrium outcomes.

LMPs decompose into an energy component, a congestion component, and a losses component [33]. In the Swedish system, which is divided into four bidding zones (SE1–SE4) reflecting north-to-south transmission constraints [9], congestion between zones creates price differences that directly affect nuclear generators' revenues. A plant's zone determines the price it receives and congestion can lead to curtailment or price discounts relative to other zones. The four-zone structure is the spatial basis for the model in Chapter 3.

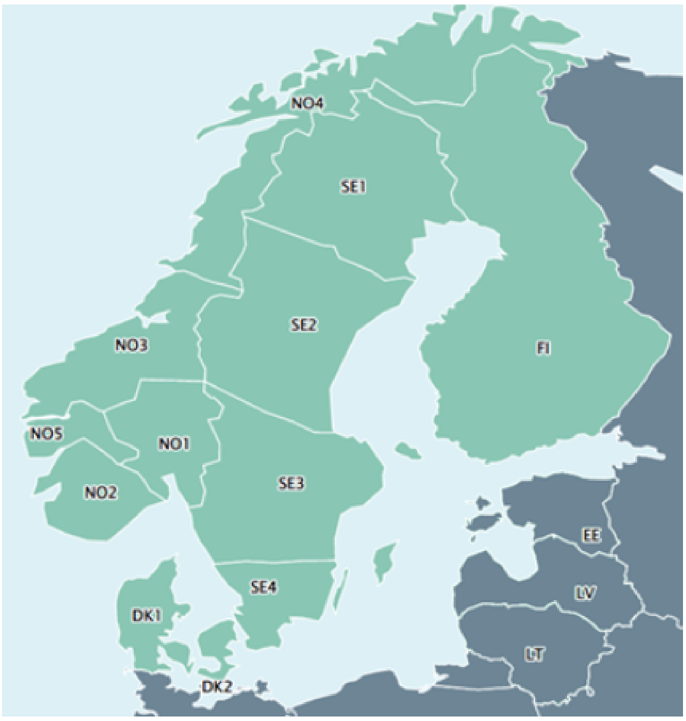

Figure 2.1: The twelve bidding zones of the Nordic power market. Source: ENTSO-E. Bidding Zone Review 2025 [9]

A key quantity throughout this thesis is the shadow price on the nuclear fuel-balance constraint. Following the hydro-reservoir analogy of Crampes and Moreaux [7], this shadow price is interpreted as the *fuel value*—the opportunity cost of consuming fuel today rather than conserving it for a higher-price future period. The fuel value plays the same role in the nuclear dispatch problem as the water value plays in hydropower scheduling, and it is the primary metric used to quantify the economic value of nuclear flexibility across scenarios.

2.2 Impact of Increased VRE Adoption on Dispatchable Plant Operations

As VRE penetration grows, the residual load—total demand minus VRE output—becomes more volatile and harder to predict. The *duck curve*, documented by California Independent System Operator [5], illustrates the canonical pattern: net load falls steeply during midday solar peaks and then ramps sharply upward in the evening, placing large and rapid flexibility demands on dispatchable resources. Under the European Green Deal [12], wind and solar capacity in Sweden and across the EU is projected to expand substantially toward 2030 and 2050 [22], making these residual-load dynamics increasingly relevant for the Swedish system.

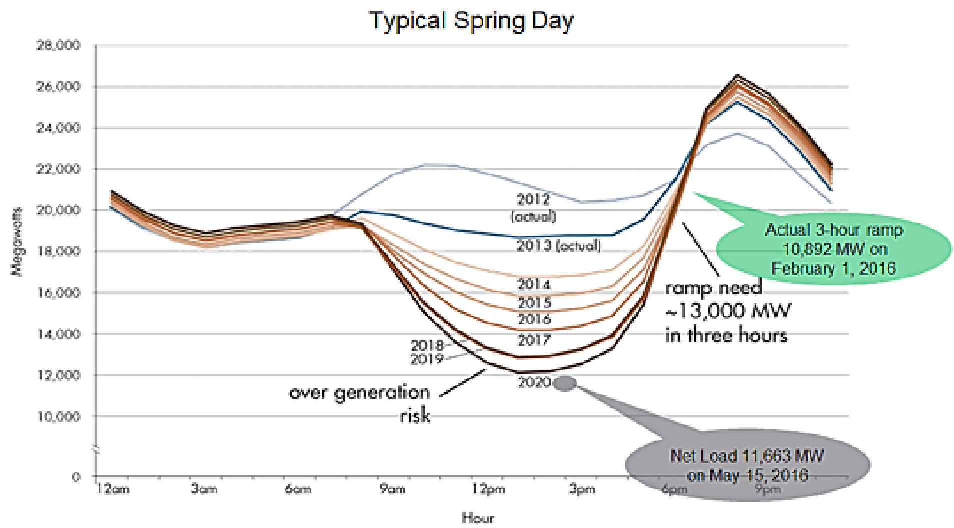

Figure 2.2: Duck Curve showing the steep ramping needs on a Typical Spring Day Source: California Independent System Operator, What the Duck Curve Tells Us about Managing a Green Grid California Independent System Operator [5]

The literature documents several consequences of high VRE penetration for nuclear plants specifically. Jenkins et al. [24] find that nuclear load-following substantially reduces system costs and VRE curtailment in high-penetration systems. Loisel et al. [27] show that load-following lowers nuclear capacity factors, affecting fixed-cost recovery, but becomes financially attractive when nuclear is not setting the marginal price. Lynch et al. [28] demonstrate that how nuclear-cycling constraints are represented materially affects optimal dispatch outcomes, motivating careful modelling of those constraints.

Sweden's generation mix makes this question particularly relevant. With hydropower supplying over 40% of electricity [10] and nuclear approximately 40% [19], the two low-carbon technologies together dominate supply. Hydropower scheduling is itself an intertemporal-stock problem: operators allocate a finite water stock across time periods to maximise value [13, 7]. Sweden's reservoirs have historically absorbed most balancing responsibility, reducing the need for nuclear load following [16]. As VRE grows, however, the question of whether hydro alone can absorb the additional variability or whether nuclear plants must shift towards active load following becomes consequential for system costs and market prices. This thesis answers that question by comparing equilibrium outcomes with and without nuclear fuel-cycle constraints under multiple VRE penetration scenarios.

2.3 Representation of Nuclear-Plant Flexibility: Ramp Rates versus Fuel Cycles

2.3.1 The Conventional Ramp-Rate Representation and Its Limitations

Standard equilibrium models represent nuclear plants as generic thermal units with capacity limits, constant marginal costs, and ramp-rate constraints [34, 32]. This implicitly treats nuclear fuel as unlimited within the modelling horizon: generating today does not reduce the ability to generate tomorrow, beyond the ramp-rate restriction. The only binding operational constraints are instantaneous capacity and the rate of output change between periods.

This representation misses two distinct physical realities. First, the frequency of output changes is constrained by reactor physics, not just their magnitude. Second, total cumulative generation over a fuel cycle is bounded by the finite energy content of the loaded fuel. Both are stock constraints rather than flow constraints, and both have economic consequences for how nuclear output should be scheduled across time.

2.3.2 Physical Basis: Xenon Dynamics and Fuel Depletion

When reactor power is reduced, xenon-135—a neutron-absorbing fission product—accumulates in the core [15]. Xenon-135 is produced both directly from fission and from the decay of iodine-135; at reduced power the lower neutron flux burns off less xenon while iodine decay continues, causing the xenon concentration to peak several hours after the power reduction [18]. This *xenon poisoning* temporarily suppresses reactivity, preventing a rapid return to full power and limiting how frequently the reactor can cycle. Thermal and mechanical stress compounds this: repeated power changes fatigue fuel claddings and structural components [18, 24], accumulating damage over the fuel cycle. The European Utilities Requirements formalise the resulting limit at two cycles per day, five per week, and 200 per year for pressurised water reactors [31]. These are hard constraints. Crucially, they bound the *number* of cycles, not just the rate of change—a stock constraint rather than a flow constraint.

Separately, the fuel loaded at the start of each 12–24 month cycle contains a fixed energy budget [18]. Total energy available is set by initial enrichment, fuel mass, and achievable burnup [15]. Generating today depletes fuel that cannot be used tomorrow. Unlike a hydro reservoir, which receives natural inflows, nuclear fuel receives no replenishment until the next refuelling outage, making fuel management a pure depletion problem analogous to the water-allocation problem of Crampes and Moreaux [7].

2.3.3 Extending the Blanchard–Massol Framework to the Swedish Context

Blanchard and Massol [3] formalise the cycling constraint as a stock problem. Using a linearised fleet-level approximation—in which the discrete per-reactor cycling budget is aggregated into a continuous fleet-level constraint with approximation error bounded by $1/N$—they embed cycling capacity as a depletable resource within a stochastic dynamic programming model of the French electricity system. Applied to a 2035 scenario, they estimate the marginal value of cycling flexibility at €100/MW at 26 cycles per year, finding that increased flexibility reduces system costs primarily by reducing solar curtailment. Notably, nuclear-operator profits decline as flexibility increases, creating a misalignment between private incentives and social optimality, i.e., operators lack the profit motive to provide flexibility even when doing so would benefit the system. [3].

Blanchard and Massol model only the cycling constraint. The fuel-energy constraint—the finite total MWh available per fuel cycle—is not part of their framework. This thesis incorporates both. The cycling constraint follows their linearised fleet-level formulation. The fuel-energy constraint is modelled as a continuously tracked depletion stock, analogous to a hydro reservoir without inflows, with its shadow price interpreted as the fuel value. Including both constraints creates a two-dimensional resource allocation problem and allows comparison of which constraint binds, and when, under different VRE penetration levels.

2.4 Research Gap

The aforementioned literature establishes two gaps that this thesis addresses.

First, the fuel-energy constraint remains largely unexamined in perfect-competition, partial-equilibrium models of electricity markets in the context of nuclear fuel cycles specifically. Standard models treat nuclear fuel as unlimited [34, 32]. Blanchard and Massol [3] advance the field by introducing the cycling-stock constraint, but their framework does not capture the intertemporal allocation of finite fuel across a refuelling cycle. This matters because the fuel-energy constraint creates an opportunity cost—the fuel value—that rises as fuel is depleted, analogous to the water value in hydro-reservoir economics [7]. Ignoring it may lead models to overestimate nuclear flexibility, underestimate VRE curtailment, and produce distorted price signals, with consequent errors in social welfare assessment.

Second, existing studies focus on the French nuclear fleet, which operates in a system with limited hydropower storage and a large nuclear share [3]. Sweden's situation is structurally different: six reactors totalling approximately 7.5 GW [19] operate alongside extensive hydropower reservoirs that have historically provided most of Sweden's flexibility [16]. The interaction between nuclear fuel-cycle constraints and hydro scheduling under high VRE penetration in a hydro-dominated system has not been studied.

This thesis bridges both gaps. A deterministic partial-equilibrium model is developed in which nuclear fuel is treated as a depletable stock analogous to a hydro reservoir, applied to the Swedish four-zone system. The deterministic approach is deliberate: the goal is to isolate the effect of the fuel-cycle representation on equilibrium outcomes, not to study uncertainty resolution—stochasticity would confound these effects and reduce tractability [6]. Scenario analysis across 2030, 2040, and 2050 VRE penetration levels quantifies the system-wide implications—for total cost, locational marginal prices, VRE curtailment, and social welfare—of incorporating the fuel-energy constraint alongside conventional ramp-rate constraints.

This work connects to the risk and decision analysis at the Department of Computer and Systems Sciences (DSV), Stockholm University. The fuel value and cycling flexibility value produced by the model are marginal values of scarce operational resources—quantities central to resource allocation under constraints, the core concern of decision analysis Chen et al.[1, 7].

3

Methodology

3.1 Choice of Research Strategy

3.1.1 Research Paradigm and Strategy Justification

Guided by a pragmatist paradigm [8], the choices of research strategy and methods in this thesis are grounded in the practical need to address the research question: how does modelling nuclear fuel cycles as a resource stock affect the assessment of nuclear flexibility and the overall equilibrium of the Swedish electricity market under high VRE penetration? This question demands a framework capable of simultaneously capturing intertemporal resource allocation, network congestion across bidding zones, and the welfare-maximising behaviour of competitive generators. The answer cannot be obtained through observation of real systems alone; instead, it requires the controlled construction and comparison of model variants in which the nuclear representation is the only element that changes.

This thesis adopts the Design-Science Research (DSR) framework as described by Johannesson and Perjons [25]. The purpose of DSR is to develop artefacts as solutions to practical problems faced by participants in a practice [25]. In the present context, the practical problem is that existing equilibrium models of electricity markets misrepresent nuclear flexibility by treating fuel availability as infinite, potentially leading to incorrect assessments of system costs, VRE curtailment, and social welfare. The artefact developed in this thesis is a partial-equilibrium model that incorporates nuclear fuel-cycle constraints—both the fuel-energy stock and the cycling-operations budget—alongside conventional generation technologies, hydropower-reservoir dynamics, and a zonal transmission network. Following the five-phase DSR process of Johannesson and Perjons [25], the thesis is structured as: (1) explicate the problem (Chapters 1–2), (2) define requirements (Chapter 3), (3) design and develop the artefact (Chapter 3), (4) demonstrate the artefact (Chapter 4), and (5) evaluate the artefact (Chapter 5).

3.1.2 Simulation as the Chosen Research Strategy

Within the DSR framework, computer simulation is adopted as the primary research strategy. Johannesson and Perjons [25] describe simulation as an imitation of reality—imitating the behaviour of a real-world system or process through an abstract mathematical model. This is precisely what this thesis does: the welfare-maximising behaviour of competitive generators, the physical constraints of the transmission network, and the depletion dynamics of nuclear fuel and hydropower reservoirs are all represented within a quadratic programming model that is solved computationally. Simulation through mathematical optimisation is a state-of-the-art strategy widely used in operations research, management science, and decision science [34, 14], and is the dominant approach in the energy economics literature addressing similar research problems, including Jenkins et al. [24], Loisel et al. [27], and Blanchard and Massol [3].

Computer simulation is particularly appropriate here because it provides a higher internal validity than field experiments: it is possible to control all variables in the environment, isolating the causal effect of the nuclear fuel-cycle representation on equilibrium outcomes [25]. The research question requires exactly this kind of controlled comparison—two model variants (Model R and Model F) that differ only in their nuclear representation, run across the same scenarios, so that any differences in outcomes can be attributed unambiguously to the change in modelling approach. As Denscombe [8] notes, computer simulation suits complex systems where field experimentation is infeasible due to scale, cost, or ethical concerns—all of which apply to the Swedish power system.

The welfare-maximising framework established by Schweppe et al. [32] and operationalised for multi-technology power systems by Chen et al. . [1] provides the theoretical foundation. Under perfect competition, maximising social welfare—the sum of consumer surplus and producer surplus net of generation costs—yields locational marginal prices as dual variables of the nodal power-balance constraints [26]. The social-welfare-maximisation problem is formulated as a quadratic programme with linear constraints, which is the natural form because linear inverse-demand functions make the consumer-surplus term quadratic while all cost and constraint expressions remain linear. This formulation additionally yields the perfectly competitive equilibrium as its solution. The resulting problem is strictly convex, guaranteeing a unique global optimum and ensuring that the Karush–Kuhn–Tucker conditions are both necessary and sufficient for optimality [4]. Strict convexity also ensures solution uniqueness, so any differences in outcomes between model variants can be attributed to the change in nuclear representation rather than to solver artefacts or multiple equilibria. The model is implemented in GAMS and solved using the CPLEX solver, which is standard practice for large-scale quadratic programming problems in energy-systems research [14].

3.1.3 Alternative Strategies Considered

Several alternative research strategies were considered and set aside on methodological grounds, following the approach recommended by Johannesson and Perjons [25] and Denscombe [8].

Experiments on live systems—Controlled experiments on live power systems are infeasible at the scale required. Artificially varying the nuclear representation of the Swedish fleet would require coordinated changes across multiple plant owners, carrying unacceptable operational risk. Simulation within a mathematical model provides the necessary controlled variation without these complications [25].

Surveys and interviews—Surveys of nuclear plant operators or market participants could yield qualitative insight into how flexibility is perceived and managed in practice [8]. However, the research question demands system-wide quantitative outcomes—total welfare, zonal prices, and curtailment volumes—that cannot be obtained from operator responses alone. Furthermore, surveys have low response rates and do not provide the agility needed to run multiple scenario comparisons [8].

Detailed case study—A case study of a single plant or bidding zone could provide granular operational detail but would sacrifice the system-wide perspective that is central to this thesis [8]. The interaction between nuclear flexibility, hydropower scheduling, VRE curtailment, and cross-zonal congestion can only be assessed at the full Swedish system level.

Stochastic programming—A stochastic formulation would capture uncertainty in VRE availability, hydropower inflows, and demand. However, the deterministic approach is deliberately chosen: the objective is to isolate the effect of the nuclear fuel-cycle representation on equilibrium outcomes, not to study the value of information or the role of uncertainty resolution. Introducing stochasticity would confound these effects and substantially increase computational cost, constraining the temporal resolution and scenario breadth achievable within the thesis scope [6].

3.2 Research Methods

3.2.1 Data-Collection

The data-collection method is based on documents from secondary sources, specifically official statistics and quantitative datasets, following Denscombe [8] and Johannesson and Perjons [25]. The dataset was compiled for the Swedish power system using 2018 as the base year and provided by the supervisor. It covers the four Swedish bidding zones (SE1-SE4) at hourly resolution over the full calendar year (8,760 time steps) and comprises the following components: hourly electricity consumption per zone [29, 10]; hourly VRE availability factors for wind and solar [10]; hourly net imports between Sweden and neighboring countries [10]; natural inflow data for run-of-river and reservoir-hydro units; and technical and economic parameters for the generation fleet and transmission network, including installed capacities, marginal costs, ramp rates, emission factors, hydro reservoir capacities and inter-zonal NTC values, sourced from ENTSO-E, Svenska Kraftnät, and the IAEA [19, 18]. The network topology and hydropower parameters follow the dataset compiled by Hassanzadeh Moghimi et al. [17], whose study of strategic reservoir operations in the Nordic system used the same Swedish zones and 2018 base year adopted here.

The inverse demand functions are calibrated to observed 2018 zonal prices and consumption following Chen et al. [1], with price elasticity set to $\varepsilon = -0.065$ and negative slopes corrected by taking the absolute value to ensure the welfare maximisation problem remains well-posed.

3.2.2 Data-Analysis Method

Data analysis refers to the process through which answers to research questions can be derived from collected data [8, 25]. Because the data at the core of this study—generation capacities, technology costs, transmission limits, and hourly demand and availability profiles—are numerical, quantitative analysis is the appropriate method [8]. Qualitative methods cannot deliver the system-wide metrics required, and statistical regression cannot extrapolate to high-VRE futures outside historical experience. Stochastic programming was rejected to isolate the causal effect of the nuclear representation without confounding uncertainty effects [6].

A welfare-maximising quadratic programming (QP) model is, therefore, developed to maximise social welfare under perfect competition subject to technical constraints representing the physical and operational limits of the Swedish power system [32, 1], implemented in GAMS and solved using the CPLEX solver [14]. The analysis proceeds in three stages: (1) **validation**, in which the baseline model is tested against 2018 historical data by comparing simulated zonal prices and generation dispatch against Nord Pool and ENTSO-E reported values; (2) **scenario analysis**, in which *Model R* (ramp-rate only) and *Model F* (fuel-cycle constraints included) are compared across VRE penetration levels representing 2030, 2040, and 2050 conditions, with outcomes measured in terms of zonal prices, VRE curtailment, and social welfare; and (3) **post-processing**, in which government carbon tax revenue is added back to recover true social welfare [1], and shadow prices on the fuel-balance and cycling-stock constraints are extracted to quantify the fuel value and flexibility value across scenarios.

3.3 Application of Research Method

This section presents the application of the quantitative simulation method described above. The mathematical formulation of the equilibrium model follows directly from the welfare-maximising framework established in Section 3.1. The nomenclature used throughout is defined first, followed by the objective function and constraints.

Nomenclature

Sets and Indices

$i \in \mathcal{I}$	Generating firms
$\ell \in \mathcal{L}$	Transmission lines
$n \in \mathcal{N}$	Network nodes
$e \in \mathcal{E}$	Variable renewable energy (VRE) types
$t \in \mathcal{T}$	Time periods
$\mathcal{T}^{ini} \subset \mathcal{T}$	Initial time periods
$u \in \mathcal{U}$	Thermal generation unit types
$u^{NU} \in \mathcal{U}^{NU} \subset \mathcal{U}$	Nuclear generation units
$h \in \mathcal{H}$	Hydro units
$\ell \in \mathcal{L}_{n^+}$	Transmission line starting at node n
$\ell \in \mathcal{L}_{n^-}$	Transmission line starting at node n
$n \in \mathcal{N}_{l^+}$	Node index for starting node of line ℓ
$n \in \mathcal{N}_{l^-}$	Node index for ending node of line ℓ
$d \in \mathcal{D}$	Days
$w \in \mathcal{W}$	Weeks
$\mathcal{T}^d \subset \mathcal{T}$	Time periods in day d
$\mathcal{T}^w \subset \mathcal{T}$	Time periods in week w

Parameters

$A^{vre}_{t,n,e}$	VRE availability factor	-
C^{max}	Maximum cycles per reactor per year	yr^{-1}
$C^{max,daily}$	Maximum cycles per reactor per day	day^{-1}
$C^{max,weekly}$	Maximum cycles per reactor per week	week^{-1}
C^{sto}	Storage operating cost	€/MWh
C^{opr}_u	Marginal operating cost	€/MWh
D	Cycle depth (fraction of capacity)	-
$D^{int}_{n,t}$	Demand function intercept	€/MWh
$D^{slp}_{n,t}$	Demand function slope	€/$(\mathrm{MWh})^2$
E^{in}	Charging efficiency	-
E^{sto}	Storage efficiency	-
$I_{i,n,t,h}$	Natural inflow to reservoir	MWh
$K^{gen}_{i,n,u}$	Installed generation capacity	MW
$K^{vre}_{n,i,e}$	VRE installed capacity	MW
K^{+}_{ℓ}	Line capacity, positive direction	MW
K^{-}_{ℓ}	Line capacity, negative direction	MW
N	Number of nuclear reactors in fleet	-
$\hat{P}_{reactor}$	Capacity per nuclear reactor	MW
P_u	CO_2 emission rate	tCO_2/MWh
$R^{ini}_{i,n,h}$	Initial storage level	MWh
$\overline{R}^{in}_{i,n,h}$	Maximum charging rate	h^{-1}
$\overline{R}^{out}_{i,n,h}$	Maximum discharging rate	h^{-1}
R^{up}_u	Ramp-up rate limit	h^{-1}
R^{down}_u	Ramp-down rate limit	h^{-1}
$\overline{R}_{i,n,h}$	Maximum storage capacity	MWh
$\underline{R}_{i,n,h}$	Minimum storage level	MWh
S	Carbon price	€/tCO_2
$S^{ini}_{i,n}$	Initial nuclear fuel stock	MWh
$\underline{S}_{i,n}$	Minimum nuclear fuel requirement	MWh
$\overline{S}_{i,n}$	Maximum nuclear fuel capacity	MWh
T_t	Length of each time period	h

Variables

$\hat{d}_t^+$	Positive variation in nuclear fleet output	MWh
$f_{\ell,t}$	Power flow on line	MW
$g_{i,n,t,e}^{vre}$	VRE generation output	MWh
$g_{i,n,t,u}$	(Thermal) generation output	MWh
$\hat{g}_t^{fleet}$	Total nuclear fleet output	MWh
m_t	Remaining nuclear flexibility stock	MWh
$q_{n,t}$	Electricity consumption	MWh
$r_{i,n,t,h}^{in}$	Energy charged into storage	MWh
$r_{i,n,t,h}^{out}$	Energy discharged from storage	MWh
$r_{i,n,t,h}^{sto}$	Storage level at end of period	MWh
$s_{i,n,t}$	Nuclear fuel remaining	MWh

Objective Function

The model maximises social welfare under perfect competition. Social welfare is represented as consumer surplus minus total generation and storage costs, inclusive of carbon tax. It should be noted that the government revenue from collecting the carbon tax needs to be added back to (3.1) in a post-processing step to reflect social welfare.

$$\max \quad \sum_{t\in T}\sum_{n\in N}\left[D_{n,t}^{int}q_{n,t} - \frac{1}{2}D_{n,t}^{slp}q_{n,t}^2 - \sum_{i\in I}\sum_{u\in U}\left(C_u^{opr} + S\cdot P_u\right)g_{i,n,t,u}\right] \tag{3.1}$$

Constraints

1. Power Balance

At each node and time period, electricity supply must equal demand, accounting for generation, consumption, network flows, and storage operations.

$$\begin{aligned} q_{n,t} - \sum_{i\in I}\sum_{u\in U} g_{i,n,t,u} - \sum_{i\in I}\sum_{e\in E} g_{i,n,t,e}^{vre} + T_t\sum_{\ell\in\mathcal{L}_n^+} f_{\ell,t} - T_t\sum_{\ell\in\mathcal{L}_n^-} f_{\ell,t} \\ - \sum_{i\in I}\sum_{h\in\mathcal{H}} r_{i,n,t,h}^{out} + \sum_{i\in I}\sum_{h\in\mathcal{H}} r_{i,n,t,h}^{in} = 0 \quad \forall n\in\mathcal{N},\ t\in\mathcal{T} \end{aligned} \tag{3.2}$$

2. Transmission-Capacity Limits

The power flow on each line is limited by its thermal capacity in both directions.

$$-K_{\ell}^{-} \leq f_{\ell,t} \leq K_{\ell}^{+} \quad \forall \ell \in \mathcal{L},\ t \in \mathcal{T} \tag{3.3}$$

3. Generation Capacity

Thermal generation cannot exceed installed capacity.

$$g_{i,n,t,u} \leq T_t \cdot K_{i,n,u}^{gen} \quad \forall i \in \mathcal{I},\ n \in \mathcal{N},\ t \in \mathcal{T},\ u \in \mathcal{U} \tag{3.4}$$

4. VRE Generation

Variable renewable generation is determined by availability factors.

$$g_{i,n,t,e}^{vre} \leq A_{t,n,e}^{vre} \cdot T_t \cdot K_{n,i,e}^{vre} \quad \forall i \in \mathcal{I},\ n \in \mathcal{N},\ t \in \mathcal{T},\ e \in \mathcal{E} \tag{3.5}$$

5. Ramping Constraints

Generation changes of non-nuclear thermal units between consecutive periods (excluding the first period) are limited by ramping capabilities.

$$g_{i,n,t,u} - g_{i,n,t-1,u} \leq T_t \cdot R_u^{up} \cdot K_{i,n,u}^{gen} \quad \forall i \in \mathcal{I},\ n \in \mathcal{N},\ t \in \mathcal{T} \setminus \mathcal{T}^{ini},\ u \in \mathcal{U} \setminus \mathcal{U}^{NU} \tag{3.6}$$

$$g_{i,n,t-1,u} - g_{i,n,t,u} \leq T_t \cdot R_u^{down} \cdot K_{i,n,u}^{gen} \quad \forall i \in \mathcal{I},\ n \in \mathcal{N},\ t \in \mathcal{T} \setminus \mathcal{T}^{ini},\ u \in \mathcal{U} \setminus \mathcal{U}^{NU} \tag{3.7}$$

6. Storage Balance

The evolution of storage levels accounts for previous storage (with efficiency losses), charging, discharging, and natural inflows. Spillage is allowed when the reservoir reaches maximum capacity.

$$r^{sto}_{i,n,t,h} \le E^{sto} \cdot R^{ini}_{i,n,h} + E^{in} \cdot r^{in}_{i,n,t,h} - r^{out}_{i,n,t,h} + I_{i,n,t,h} \quad \forall i, n, t \in \mathcal{T}^{ini}, h \tag{3.8}$$

$$r^{sto}_{i,n,t,h} \le E^{sto} \cdot r^{sto}_{i,n,t-1,h} + E^{in} \cdot r^{in}_{i,n,t,h} - r^{out}_{i,n,t,h} + I_{i,n,t,h} \quad \forall i, n, t \in \mathcal{T} \setminus \mathcal{T}^{ini}, h \tag{3.9}$$

7. Storage Limits

Storage levels must remain within physical bounds.

$$r^{sto}_{i,n,t,h} \le \overline{R}_{i,n,h} \quad \forall i, n, t, h \tag{3.10}$$

$$r^{sto}_{i,n,t,h} \ge \underline{R}_{i,n,h} \quad \forall i, n, t, h \tag{3.11}$$

8. Charging and Discharging Rate Limits

Storage charging and discharging rates are limited by equipment capacity.

$$r^{in}_{i,n,t,h} \le T_t \cdot \overline{R}^{in}_{i,n,h} \cdot \overline{R}_{i,n,h} \quad \forall i, n, t, h \tag{3.12}$$

$$r^{out}_{i,n,t,h} \le T_t \cdot \overline{R}^{out}_{i,n,h} \cdot \overline{R}_{i,n,h} \quad \forall i, n, t, h \tag{3.13}$$

9. Nuclear-Fuel Balance

Nuclear fuel is treated as a depletable resource, similar to a hydro reservoir. The fuel remaining at the end of each period equals the previous fuel minus the total energy generated by all nuclear units at that location.

$$s_{i,n,t} = S^{ini}_{i,n} - T_t \cdot \sum_{u \in \mathcal{U}^{NU}} g_{i,n,t,u} \quad \forall i, n,\ t \in \mathcal{T}^{ini} \tag{3.14}$$

$$s_{i,n,t} = s_{i,n,t-1} - T_t \cdot \sum_{u \in \mathcal{U}^{NU}} g_{i,n,t,u} \quad \forall i, n,\ t \in \mathcal{T} \setminus \mathcal{T}^{ini} \tag{3.15}$$

10. Nuclear-Fuel Limits

Storage levels must remain within physical bounds.

$$s_{i,n,t} \geq \underline{S}_{i,n} \quad \forall i, n,\ t \in \mathcal{T} \tag{3.16}$$

$$s_{i,n,t} \leq \overline{S}_{i,n} \quad \forall i, n,\ t \in \mathcal{T} \tag{3.17}$$

11. Nuclear-Fleet Output

To implement fleet-level cycle constraints, total nuclear generation is defined as:

$$\hat{g}_t^{fleet} = \sum_{i \in \mathcal{I}} \sum_{n \in \mathcal{N}} \sum_{u \in \mathcal{U}^{NU}} g_{i,n,t,u} \quad \forall t \in \mathcal{T} \tag{3.18}$$

12. Positive Variation Definition

Following Blanchard and Massol (2025), we track only positive changes in fleet output, as each load-following cycle consists of a reduction followed by an increase. The positive variation variable captures the upward movement that completes cycles.

$$\hat{d}_t^{+} \geq \hat{g}_t^{fleet} - \hat{g}_{t-1}^{fleet} \quad \forall t \in \mathcal{T} \setminus \mathcal{T}^{ini} \tag{3.19}$$

$$\hat{d}_t^{+} \geq 0 \quad \forall t \in \mathcal{T} \tag{3.20}$$

$$\hat{d}_t^{+} = 0 \quad \forall t \in \mathcal{T}^{ini} \tag{3.21}$$

13. Flexibility-Stock Evolution

The remaining flexibility (cycle capacity) is tracked as a stock that depletes with positive variations.

$$m_0 = N \cdot C^{max} \cdot D \cdot \hat{P}_{reactor} \tag{3.22}$$

$$m_t = m_{t-1} - \hat{d}_t^{+} \quad \forall t \in \mathcal{T} \tag{3.23}$$

$$m_t \geq 0 \quad \forall t \in \mathcal{T} \tag{3.24}$$

14. Multi-Timescale Cycle Limits

To prevent excessive wear and maintain safe operation, nuclear reactors face cycle limits at multiple timescales.

Daily Limit:

$$\sum_{t \in \mathcal{T}^d} \hat{d}_t^+ \leq N \cdot C^{max,daily} \cdot D \cdot \hat{P}_{reactor} \quad \forall d \tag{3.25}$$

Weekly Limit:

$$\sum_{t \in \mathcal{T}^w} \hat{d}_t^+ \leq N \cdot C^{max,weekly} \cdot D \cdot \hat{P}_{reactor} \quad \forall w \tag{3.26}$$

Annual Limit (Stock Constraint):

$$\sum_{t \in \mathcal{T}} \hat{d}_t^+ \leq N \cdot C^{max} \cdot D \cdot \hat{P}_{reactor} \tag{3.27}$$

15. Non-Negativity Constraints

$$\begin{aligned}
&g_{i,n,t,u} \geq 0, && \forall i \in \mathcal{I}, n \in \mathcal{N}, t \in \mathcal{T}, u \in \mathcal{U} \\
&g^{vre}_{i,n,t,e} \geq 0, && \forall i \in \mathcal{I}, n \in \mathcal{N}, t \in \mathcal{T}, e \in \mathcal{E} \\
&q_{n,t} \geq 0, && \forall n \in \mathcal{N}, t \in \mathcal{T} \\
&r^{in}_{i,n,t,h} \geq 0, && \forall i \in \mathcal{I}, n \in \mathcal{N}, t \in \mathcal{T}, h \in \mathcal{H} \\
&r^{out}_{i,n,t,h} \geq 0, && \forall i \in \mathcal{I}, n \in \mathcal{N}, t \in \mathcal{T}, h \in \mathcal{H} \\
&r^{sto}_{i,n,t,h} \geq 0, && \forall i \in \mathcal{I}, n \in \mathcal{N}, t \in \mathcal{T}, h \in \mathcal{H} \\
&s_{i,n,t} \geq 0, && \forall i \in \mathcal{I}, n \in \mathcal{N}, t \in \mathcal{T} \\
&m_t \geq 0, && \forall t \in \mathcal{T} \\
&\hat{g}^{fleet}_t \geq 0, && \forall t \in \mathcal{T}
\end{aligned}$$

3.4 Research Ethics

This study involves no human subjects or personal data. All data are sourced from publicly available datasets and official statistics, with no confidential or sensitive information involved. The research procedures comply with Swedish legislation and the ethical guidelines of Stockholm University [25, 8].

No ethical approval from an ethics committee is required for this thesis, as no data from any human or animal subject were collected or used. Data on electricity demand are aggregated zonal statistics, and data on electricity generation come from producers' own publicly reported figures and from official databases such as Nord Pool and ENTSO-E. No personal or sensitive personal data are involved at any stage of the research.

In the interest of transparency and replicability, the complete mathematical formulation is provided in Section 3.3 and the GAMS implementation is included in Appendix B. All data sources are clearly cited and all modeling assumptions are explicitly stated, enabling independent reproduction of the results.

3.5 Use of AI and Software

3.5.1 Software

The equilibrium model is implemented in the General Algebraic Modeling System (GAMS, version 41.5.0) and solved using the CPLEX solver (version 22.1), which is standard practice for large-scale quadratic-programming problems in energy-systems research [14]. GAMS provides a high-level algebraic modelling language that facilitates the transparent specification of the objective function, constraints, and scenario parameters, and the CPLEX solver exploits the convex quadratic structure of the problem to guarantee globally optimal solutions efficiently. Data pre-processing and result visualisation are performed in Python 3.11 using the `pandas`, `numpy`, and `matplotlib` libraries.

3.5.2 Use of Artificial Intelligence

Artificial-intelligence tools, specifically large language model (LLM) assistants, were used during the preparation of this thesis for the following purposes: drafting and editing written text, checking grammar and academic style, and assisting with LaTeX formatting. All substantive intellectual contributions—the research design, model formulation, mathematical derivations, data analysis, interpretation of results, and academic judgements—are the original work of the authors. AI-generated text was always reviewed, verified, and revised by the authors before inclusion. The use of AI tools complies with the guidelines of Stockholm University regarding the responsible use of AI in academic work.

4

Results

5

Discussion

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